MUSKAN SANCHETI

COMPUTER SCIENCE ENGINEER

9829012362 | LINKEDIN | GITHUB | muskansancheti5@gmail.com

PROFESSIONAL SUMMARY

Computer Science undergraduate with strong foundations in DSA, C++, Python, and JavaScript, with experience building software applications, machine-learning pipelines, and real-time graphics systems. Interested in software engineering, scalable systems, development, and database-driven applications. Experienced in developing projects involving Flask, WebGL2, OpenCV, and data processing.

PROJECTS

Hostel Management System (React, JavaScript, SQL) Ongoing

- Developing a full-stack web application for university hostel management with modules for student records, room allocation, complaints, attendance, and hostel administration than to connect the frontend with the backend and database.

AI Technical PDF Translator (Next.js, Flask, OCR, NLP) July-2026

- Developed an end-to-end Slovak-to-English technical PDF translation system integrating OCR, machine translation, custom terminology handling, and layout-aware PDF reconstruction.
- Built backend workflows to process PDF documents and coordinate OCR, translation, and document reconstruction.

Manufacturing Process Capability Analysis & Cpk Prediction (Python, ML) May-July'26

- Developed a process capability analysis system to calculate Cp, Cpk, Pp, and Ppk and generate histograms, control charts, and probability plots from manufacturing data.
- Built a Random Forest regression pipeline to predict next-day process capability using historical manufacturing data.

Real-Time Ray Tracing Engine (WebGL2, GLSL) Dec-2025

- Developed a GPU-based ray tracing engine implementing ray-sphere intersection, shadow rays, recursive reflections, and Phong shading.
- Implemented real-time rendering through fragment-shader computations and interactive camera movement.

AI-Based Fitness Training System (Python, OpenCV, MediaPipe) June-2025

- Developed a computer-vision based fitness application that tracks body posture in real-time using a webcam.

EXPERIENCE

Data Analytics and AI Intern

NBC Bearings (National Engineering Industries Ltd.) | Jaipur, Rajasthan

- Developed a Python-based process capability analysis pipeline to calculate Cp, Cpk, Pp and Ppk from manufacturing data and generate statistical visualizations.
- Built a Flask-based application for automated process capability analysis, including histogram, control-chart and probability-plot generation.
- Developed a Random Forest regression pipeline to predict next-day process capability using historical manufacturing data.
- Automated PDF translation workflows using OCR, NLP and machine translation for technical documents.

SKILLS

- **Programming Languages:** Python, C, C++, JavaScript, SQL
- **Core Competencies:** Machine Learning, Data Structures & Algorithms, Computer Graphics, Linear Algebra, Computer Networks, Operating Systems, DBMS
- **Web & Backend**: HTML, CSS, JavaScript, React, Flask
- **AI / ML**: Scikit-learn, Hugging Face Transformers, EasyOCR, OpenCV
- **Graphics & Tools**: WebGL2, GLSL, Git, GitHub, VS Code

EDUCATION

JK LAKSHMIPAT UNIVERSITY 2024-2028
BTech - Computer Science | **CGPA: 9.6**

IIT KANPUR 2025
Semester Exchange

MAHAVEER PUBLIC SCHOOL 2012-2024
High school
